

TURBULENT PRANDTL NUMBER IN THE NEAR-WALL REGION OF THERMAL CONVECTION

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Abstract

Prandtl number alters the behaviour of convective flows, and an understanding of the ways in which it does so is required to model flows in various natural and industrial settings. The turbulent Prandtl number — a characteristic of a turbulent flow — is needed in various turbulence closure models. In this paper, we study how the turbulent Prandtl number in regions near the horizontal plates of a convective flow is affected by the molecular Prandtl number Pr . We perform direct numerical simulations of Rayleigh-Bénard convection in a two-dimensional square box and find that the mean turbulent Prandtl number in the thermal boundary layer region remains smaller than one, and does not follow any specific trend with respect to its molecular counterpart. We also utilize the machine learning methods to understand the variation of the Nusselt number with respect to the governing parameters and find that the random forest regressor satisfactorily captures the trend in 2D RBC.

Keywords: Turbulent convection, Direct numerical simulations, Turbulence modelling

1 INTRODUCTION

Rayleigh-Bénard convection (RBC) is a sought-after model to explore turbulent flows driven by temperature inhomogeneities [38, 37]. Examples of such flows include those in interior as well as atmosphere of various celestial bodies, including Earth. The effective thermal driving, at which a convective flow operates is quantified using the Rayleigh number Ra , which together with the Prandtl number Pr controls the behaviour of convective flows. As convective flows in various natural as well as industrial settings operate at drastically different Prandtl numbers (e.g., $Pr \sim 10^{-6}$ in the solar convection zone, $Pr \sim 10^{-2}$

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in the Earth’s outer core, and $Pr \sim 10^{23}$ in the Earth’s mantle), a better understanding of the influence of Pr on various flow characteristics is required. Note that Pr is related to the strength of inertial forces in a flow; low- Pr flows, being highly inertial, exhibit differing characteristics than their corresponding high- Pr counterparts do [27, 17, 26]. It is, however, challenging to systematically cover a wide range of Prandtl numbers in experimental investigations of RBC as different working fluids are required for this purpose. Direct numerical simulations (DNS) facilitate this exploration but are limited to moderate Rayleigh numbers as computational demand grows significantly when $Pr \ll 1$ [28, 6].

To achieve a wider range of Prandtl numbers, spatially-confined convective flows are commonly considered because the needed computational resources scale as Γ^2 , where the aspect ratio Γ is defined as the ratio of the horizontal and vertical dimensions of the convection domain. Pandey and Sreenivasan [15] and Pandey et al. [20] performed DNS in a slender cylinder of diameter one-tenth its height for Pr between 0.005 and 200 and Ra up to 3×10^{10} , and found that the Pr -dependence of the global heat and momentum transports as well as temperature fluctuations and their distribution are similar to those in horizontally-extended domains, particularly when the thermal driving is strong. Two-dimensional (2D) models of convection are also frequently utilized for purposes of exploring wider ranges of governing parameters [24, 5, 35]. Though 2D RBC is fundamentally different from its fully three-dimensional (3D) counterpart, both share some commonalities. In this work, we therefore consider convection in a closed 2D square box driven strongly at $Ra = 10^9$ and explore the Pr -dependence of large-scale flow organization, the global heat and momentum transports, and their fluctuations.

Large-scale flow organization affects the properties of a convective flow. Generally, a single large-scale structure — rotating in the clockwise or counter-clockwise direction — is established in a 2D square box with the spatial velocity distribution depending on the governing parameters: mean domain-sized structure is circular/squarish at low Prandtl numbers [14], whereas oriented along one of the diagonals at moderately-high Prandtl numbers [3]. By systematically varying Pr , we observe in this work that the mean flow organization evolves from a vortex-like structure at low Prandtl numbers to diagonally-oriented structure at moderate Prandtl numbers to squarish structure at high Prandtl numbers. This evolution affects the near-wall variation of velocity and its gradients, and, in turn, the quantities depending on these gradients [12]. One such quantity is the turbulent (or effective) viscosity ν_t , defined as the ratio of the Reynolds stress and the mean velocity gradient. We observe that the mean velocity gradient flips sign in the near-wall region for moderately large Prandtl numbers and consequently hinders the estimation of ν_t , see Sec. 3.

Note that mixing is more efficient in turbulent flows than in laminar flows, indicating that the turbulent diffusion coefficients far exceed their molecular counterparts [12]. Mixing length arguments enable us to estimate these coefficients. In addition to ν_t , convective flows are characterized by turbulent thermal diffusivity κ_t , which is defined, in the flux-gradient (FG) framework, as a ratio of the turbulent heat flux and the mean temperature gradient [10]. In RBC,

however, owing to the absence of large-scale mean gradients [25], estimating ν_t and κ_t within the FG framework is not so straightforward. (Hereafter, we jointly refer to ν_t and κ_t as the turbulent diffusivities.) Given the importance of these diffusivities, other methods are therefore sought, such as the $k-\epsilon$ method [19, 20] or a framework to compute scale-resolved turbulent diffusivities based on the Kolmogorov’s renormalized similarity hypothesis [1]. In the vicinity of the horizontal plates in RBC, boundary layers with nearly well-defined gradients are formed, making this region suitable for applying the FG framework [34].

Turbulent Prandtl number, $Pr_t = \nu_t/\kappa_t$, characterizes the fluctuations of heat and momentum in strongly-driven convective flows. It has been found that Pr_t exceeds unity in low- Pr flows [9, 2, 34], though the Reynolds analogy suggests $Pr_t \approx 1$ [12]. A high effective Prandtl number in low- Pr flows, such as that in the Sun, has been argued to provide an explanation for the convective conundrum in the Sun [8], where convective flows with significantly lower velocities are able to transport large amounts of thermal energy [19]. In this work, we apply the FG framework in the thermal boundary layer (TBL) region to estimate Pr_t , and find, however, that the trend with respect to Pr remains inconclusive, as Pr_t lies between 0.6 and 0.85 when Pr varies from 0.021 to 1.

The Nusselt number Nu and the Reynolds number Re quantify, respectively, the global heat and momentum transports in a convective flow, and exhibit power law dependences [38]: $Nu \sim Pr^\gamma$ and $Re \sim Pr^\zeta$. Nu in 3D RBC generally varies as $Pr^{1/7}$ for $Pr \lesssim 0.3$, and becomes nearly insensitive to or a weakly decreasing function of Pr at higher Prandtl numbers [7, 11]. The Reynolds number has been observed to decrease as $Pr^{-2/3}$ at low Prandtl numbers [21], whereas as Pr^{-1} at high Prandtl numbers [15]. However, available data from 2D RBC literature for $Ra \leq 10^8$ have not exhibited analogous behaviours [36, 39]. Particularly, the Reynolds number has been observed to follow a different trend, with [13] suggesting that $Re \sim Pr^{-1}$ in 2D turbulent convection.

In this paper, we study RBC in a 2D domain of $\Gamma = 1$ by conducting DNS for a high Rayleigh number $Ra = 10^9$ and Prandtl numbers varying from 0.021 to 50. The simulation parameters are briefly described in Sec. 2. We systematically explore the effects of Pr on the large-scale flow organization and global transports in Sec. 3, and on the turbulent Prandtl number in Sec. 4. A summary of the main findings of this work is presented in Sec. 5.

2 SIMULATION PARAMETERS

We perform DNS to solve the following governing equations under the Boussinesq approximation :

$$\nabla \cdot \mathbf{u} = 0, \quad (1)$$

$$\frac{\partial \mathbf{u}}{\partial t} + \mathbf{u} \cdot \nabla \mathbf{u} = -\nabla p + T \hat{\mathbf{z}} + \sqrt{\frac{Pr}{Ra}} \nabla^2 \mathbf{u}, \quad (2)$$

$$\frac{\partial T}{\partial t} + \mathbf{u} \cdot \nabla T = \frac{1}{\sqrt{RaPr}} \nabla^2 T. \quad (3)$$

Pr	$N_e N^2$	Nu	Re	$t_{sim}(t_f)$
0.021	4014^2	37.43 ± 28	443526 ± 53505	159
0.05	2478^2	41.51 ± 27	185128 ± 30221	250
0.1	2070^2	42.72 ± 24	95499 ± 15598	304
0.2	1150^2	45.98 ± 24	46456 ± 9344	519
0.35	1150^2	48.10 ± 26	27473 ± 6068	530
0.5	900^2	48.64 ± 32	20492 ± 3885	551
0.7	690^2	49.22 ± 34	15499 ± 3374	888
1	700^2	50.87 ± 35	12012 ± 3324	2112
2	690^2	51.09 ± 12	5240 ± 822	3937
4.38	690^2	52.67 ± 12	2698 ± 543	1971
7	690^2	52.27 ± 13	1761 ± 366	2318
20	690^2	53.41 ± 11	700 ± 184	3654
50	690^2	52.12 ± 11	320 ± 90	5211

Table 1: Parameters of simulations in a closed 2D square box for a fixed $Ra = 10^9$. $N_e N^2$ is the total number of mesh cells used to decompose the simulation domain, and t_{sim} is the simulation time (in free-fall units) after the flow enters a statistically steady state. Error bars in Nu and Re indicate the standard deviation.

Here $\mathbf{u} = (u_x, u_z)$, p , T represent velocity, pressure, and temperature fields, respectively. The equations are written in dimensionless units using the free-fall velocity $u_f = \sqrt{\alpha g \Delta T H}$, cell height H , free-fall time $t_f = H/u_f$, and the temperature difference ΔT between bottom and top plates. The parameters $Ra = \alpha g \Delta T H^3 / (\nu \kappa)$ and $Pr = \nu / \kappa$, where α , ν , and κ are the thermal expansion coefficient, kinematic viscosity, and thermal diffusivity of the fluid, respectively, and g is the acceleration due to gravity. The equations are integrated in time using a spectral element solver NEK5000 [4], where the flow domain is decomposed into N_e elements, and further resolved using a Legendre interpolation polynomial of order N . Thus, there are a total of $N_e N^2$ mesh cells in the entire domain.

In the present study, we vary Pr from 0.021 to 50 at a fixed $Ra = 10^9$ in a closed box of length $L = H$. The horizontal walls are isothermal with the bottom maintained at a higher temperature than the top, and the sidewalls are adiabatic. No-slip condition is maintained on all the walls. All the simulations are initialized from the conduction state. Note that the transient time needed for a 2D convective flow to reach a statistically steady state is very long [13], especially for low- Pr flows, as the transient time increases linearly with Re [16, 22]. Important details of the simulations are presented in table 1, and the analyses are conducted only after the flow achieves a statistically steady state.

3 FLOW ORGANIZATION AND GLOBAL RESPONSE PARAMETERS

3.1 Flow organization

The averaged velocity distribution is shown in figure 1 for three decades of Prandtl number. The depicted velocity field is defined as $\sqrt{\overline{u_x}^2 + \overline{u_z}^2}$, where $\overline{u_x}$ and $\overline{u_z}$ are the components of the velocity field averaged over the entire simulation time. We observe a systematic difference in the flow structure as Pr increases. The mean velocity magnitude exhibits a minimum at the center, while the surrounding flow traces approximately a closed, circular path — a structure qualitatively resembling a vortex for $Pr = 0.021$ (figure 1(a)). As Pr increases, the flow organization changes: the velocity magnitude decreases, but the spatial extent covered by the large-scale flow increases, and the flow structure begins to deform from the perfect circular shape as shown in figure 1(b,c). A qualitatively similar flow organization persists up to $Pr = 1$; see also [31] for further discussion on velocity distribution in 2D RBC. Flow for $Pr = 2$ exhibits a very different morphology: a diagonally oriented large-scale circulation with considerably strong corner rolls is formed, as shown in figure 1(d). A similar flow pattern is also observed for $Pr = 4.38$ and $Pr = 7$ (figure 1(e)); see also [3]. The presence of strong corner flow structures significantly weakens the mean velocity near the horizontal plates because the horizontal velocity changes sign as one moves from the central region toward the corner regions. This hinders the development of a mean velocity gradient in the TBL region. For $Pr = 50$ (figure 1(f)) — and also for $Pr = 20$ — the large-scale flow travels closer to the walls, forming a near-square flow pattern that covers the entire flow domain.

The horizontally- and time-averaged horizontal velocity $\langle u_x \rangle_{L,t}$ is shown as a function of height in figure 2. It can be seen that the velocity profile changes sign in the bulk of the flow due to the large-scale circulation. Hence, there is no system-scale velocity gradient in RBC; see also Samuel et al. [25]. However, as $\langle u_x \rangle_{L,t}$ does not change sign in the vicinity of the walls except for $Pr = 2, 4.38$ and 7 , it may be possible to compute the turbulent viscosity using the flux-gradient method in this region. For $Pr = 2, 4.38$ and 7 , the presence of strong corner rolls causes $\langle u_x \rangle_{L,t}$ to change sign also in the near-wall region (figure 2(b)), making it difficult to estimate the turbulent viscosity. We therefore restrict the discussion of Pr_t in Sec. 4.2 up to $Pr = 1$.

3.2 Global transports of heat and momentum

Convective state enhances the heat transport relative to that in the conductive state. Heat transfer response generally varies with the molecular properties of the fluid under a prescribed thermal forcing. The convective heat transfer is quantified by the Nusselt number, which is a ratio of the total heat flux to that due to conduction alone, and computed as

$$Nu = 1 + \sqrt{RaPr} \langle u_z T \rangle_{V,t}, \quad (4)$$

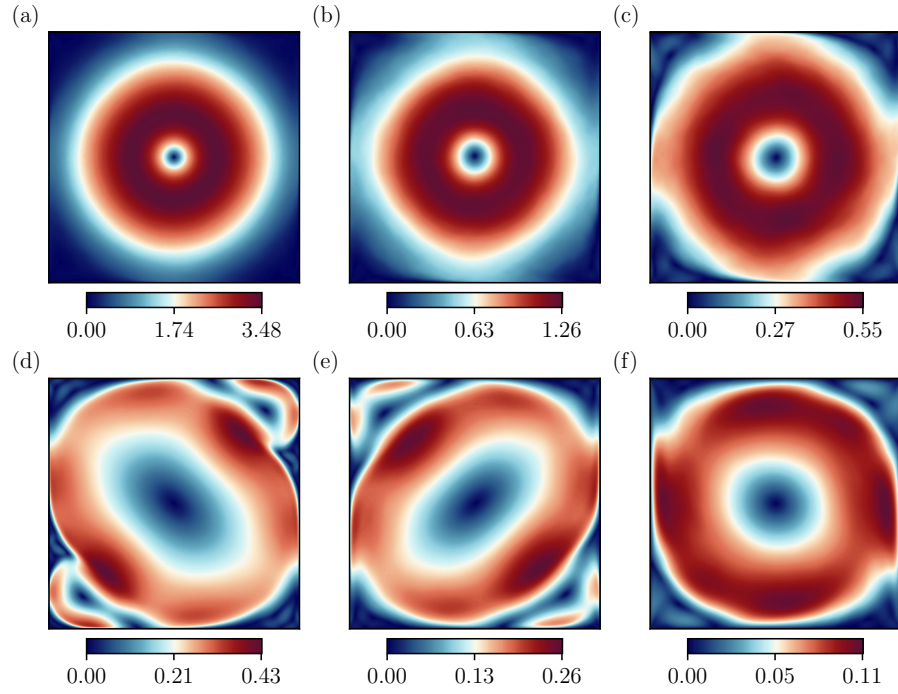


Figure 1: Mean flow structures depicted by contours of the velocity magnitude for (a) $Pr = 0.021$, (b) $Pr = 0.1$, (c) $Pr = 0.35$, (d) $Pr = 2$, (e) $Pr = 7$, (f) $Pr = 50$. The flow consists of a single large-scale structure, but the morphology evolves with Pr . Particularly for $Pr = 2$ and 7 , a large-scale circulation oriented along one of the diagonals is established.

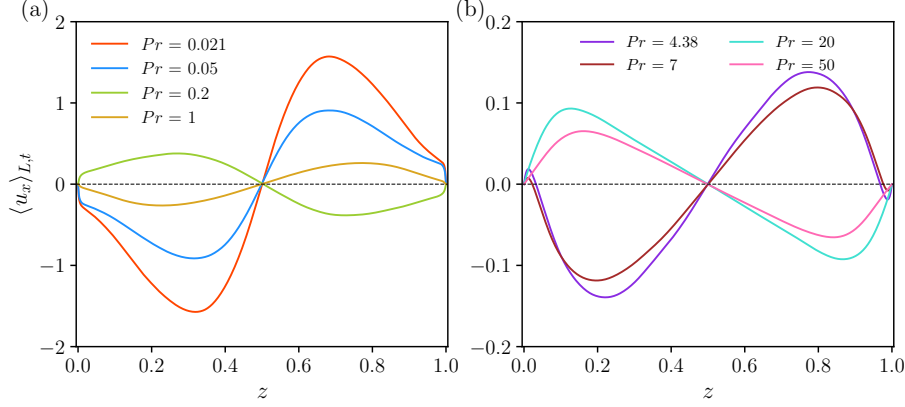


Figure 2: Vertical profiles of the mean horizontal velocity for (a) $Pr \leq 1$, and (b) $Pr > 1$. The horizontal velocity changes sign in the mid-horizontal plane, and a large-scale gradient is therefore absent. In the near-wall region, a mean gradient is established. However, for $Pr = 4.38$ and 7 , the mean velocity changes sign in this zone, making a well-defined gradient difficult to identify.

where $\langle \cdot \rangle_{V,t}$ denotes averaging over the entire flow domain and simulation time in the statistically steady state. Nu could also be obtained from the dissipation rates [32], and the convergence of Nu across all methods suggests that the simulation is sufficiently well-resolved [33, 16]. We have confirmed the convergence of Nu across all the methods and found agreement to within a few percent.

Figure 3(a) shows the variation of Nu with Pr : Nu follows an increasing trend up to $Pr \approx 4.38$, and then exhibits a weaker variation about a mean. The best fit up to $Pr \approx 1$ results in a scaling exponent of $\gamma = 0.08$. Note that a higher exponent $\gamma = 0.11$ has been observed at $Ra = 10^8$ for $0.01 \leq Pr \leq 0.15$, and $\gamma = 0.14$ at $Ra = 10^7$ and $0.005 \leq Pr \leq 0.07$ by [39] in an identical domain. These observations indicate that Nu becomes less sensitive to Pr as the thermal forcing increases. The observed variation of Nu at low Prandtl numbers agrees qualitatively with the trend in 3D [7, 15]. However, the exponent γ in 2D differs and seems to evolve with Ra . Further, in contrast to that in 3D, our data at higher Pr do not follow any systematic trend.

The momentum transport is quantified by the Reynolds number, which characterizes the strength of the inertial force relative to the viscous force in the flow. We compute it as

$$Re = \sqrt{Ra/Pr} u_{RMS}, \quad (5)$$

where $u_{RMS} = \sqrt{\langle u_x^2 + u_z^2 \rangle_{V,t}}$ stands for the root mean square (RMS) velocity. Figure 3(b) shows Re as a function of Pr , where two scaling regimes are observed: $Pr > 1$ corresponds to $\zeta = -0.87$, while $Pr \leq 1$ to $\zeta = -0.95$. Zhang and Zhou [39] found the exponent $\zeta = -0.93$ at $Ra = 10^8$, while $\zeta = -0.82$ at $Ra = 10^7$, indicating that Re decreases with Pr more rapidly as the driving

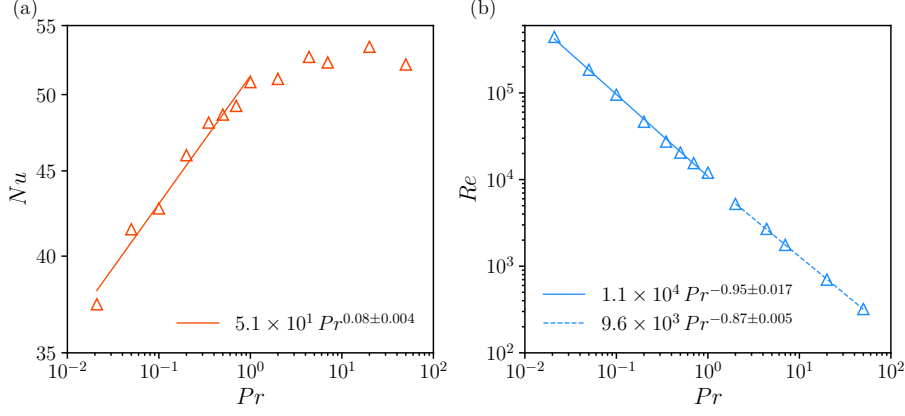


Figure 3: Global transports versus Pr for $Ra = 10^9$. (a) Nu increases up to $Pr \approx 4.38$, and becomes nearly unchanging thereafter. A best fit of the data up to $Pr = 1$ results in $Nu \sim Pr^{0.08}$. (b) Re decreases monotonically but shows two different scalings: best fits yield scaling exponents of -0.95 and -0.87 for $Pr \leq 1$ and $Pr > 1$, respectively.

force increases. Lindborg [13] suggests that Re in 2D RBC varies as Pr^{-1} . Therefore, the present results, along with those discussed in the literature, indicate that the Pr^{-1} scaling is approached progressively as the thermal driving becomes stronger.

3.3 Forecasting heat transport using machine learning

The findings of the present work together with those from literature suggest that the scalings of the global heat and momentum transports in 2D RBC differ in subtle ways than those in 3D RBC. However, in contrast to various models available in 3D RBC to understand the scalings of Nu and Re [32, 7, 17, 18], no dedicated model exists to capture the variation of the global quantities in two dimensions. With the advent of the machine learning (ML) methods [23], it may be possible to build a data-driven model and forecast these crucial parameters. We discuss one such model to predict the Nusselt number in a 2D square box in the following.

To this end, we utilize heat transport data from the references [36, 14, 39, 16, 22] together with that in the present work. We primarily chose data where Pr is varied at a fixed Ra . The dataset contains a total of 55 data points (obtained from DNS in a 2D square box) and covers more than four orders of magnitude in Prandtl number ($0.005 \leq Pr \leq 55$) and three orders in Rayleigh number ($10^6 \leq Ra \leq 10^9$). With an aim to build a regression model that accepts Prandtl and Rayleigh numbers as input parameters of a convective flow in a closed square box and yields the Nusselt number as an output parameter, we tried various

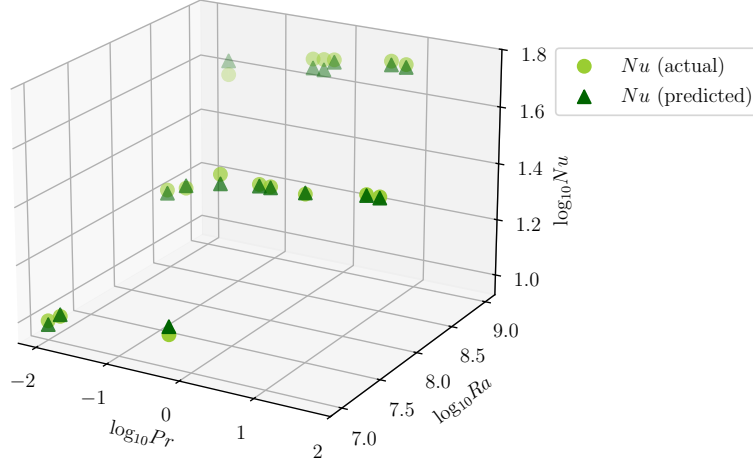


Figure 4: Comparison of the predicted and actual heat transport efficiencies in 2D RBC in a closed square box. The Nusselt number is forecasted with a random forest regression model and the overall performance of the model is excellent as the R^2 -score is 0.98.

models, such as the multiple linear regression, polynomial regression, support vector regression, etc. We found, however, that the random forest regression model suits best for our purposes, and therefore we proceed with this model.

To train the random forest regressor, we randomly select 70% of the available data and use the trained model to predict the heat transport for rest 30% of the data. The total number of decision trees — a hyperparameter of the model — is chosen to be five. Note, however, that on varying this parameter between 5 and 100, we found no significant difference in the performance of the model. The predicted Nusselt numbers on the test set are compared against the actual values in figure 4, which exhibits that the regressor is able to capture the underlying trend quite well. The predicted Nusselt numbers (dark green triangles) agree closely with the corresponding values from DNS (light green circles) for all the cases. To quantify the efficiency of the random forest regressor, we compute the coefficient of determination, known as the R^2 -score. We find that the regressor yields an R^2 -score of 0.98 against the test set, indicating that the overall prediction of the global heat transport from the regressor is excellent.

Note that we also tried to build a similar model to predict the Reynolds number in 2D RBC. However, due to a relatively smaller size of the available dataset, a reliable regression model could not be built. Therefore, we plan to take up this task in future when more data is available to efficiently train the model.

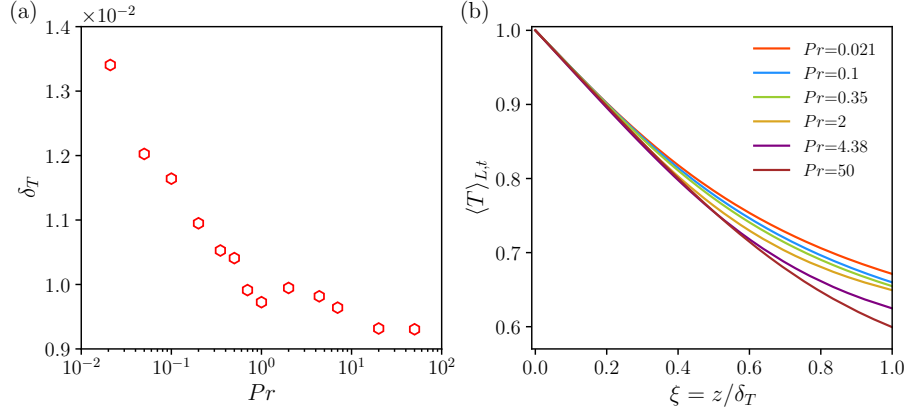


Figure 5: (a) Variation of TBL thickness with Pr . Consistent with the variation of Nu , δ_T decreases monotonically up to $Pr \approx 1$. (b) Mean temperature profiles as a function of the normalized distance, $\xi = z/\delta_T$, within the TBL region. The mean temperature drops consistently, establishing a well-defined gradient in this region.

4 FLOW FLUCTUATIONS AND TURBULENT PRANDTL NUMBER

Since a mean velocity gradient exists in the near-wall region, we need to identify a region near the horizontal walls where turbulent diffusivities could be estimated reliably. For $Pr \leq 1$, the viscous boundary layer is thinner than the thermal boundary layer. To cover a broader region near the walls, we therefore adopt the TBL thickness as the boundary of the near-wall region to compute the turbulent diffusivities. We compute the TBL thickness δ_T using the slope method [14] and show its variation with Pr in figure 5(a). The thickness decreases monotonically up to $Pr = 1$, and tends to decrease again after a slight increase at $Pr = 2$ [39]. Note that δ_T could also be estimated as $0.5/Nu$, which implies that the extent of the diffusion dominated region decreases when the heat transport increases. We have confirmed that the thicknesses computed using both the methods agree well.

To see the existence of a mean temperature gradient in the near-wall region, we plot $\langle T \rangle_{L,t}$ as a function of the normalized distance, $\xi = z/\delta_T$, from the wall, as shown in figure 5(b). The approach to the bulk temperature is gradual for low Prandtl numbers and becomes increasingly rapid as Pr increases. This feature is similar to that observed in a slender cell for $0.005 \leq Pr \leq 200$ [15]. The monotonic temperature drop within the TBL region implies that a mean temperature gradient exists there.

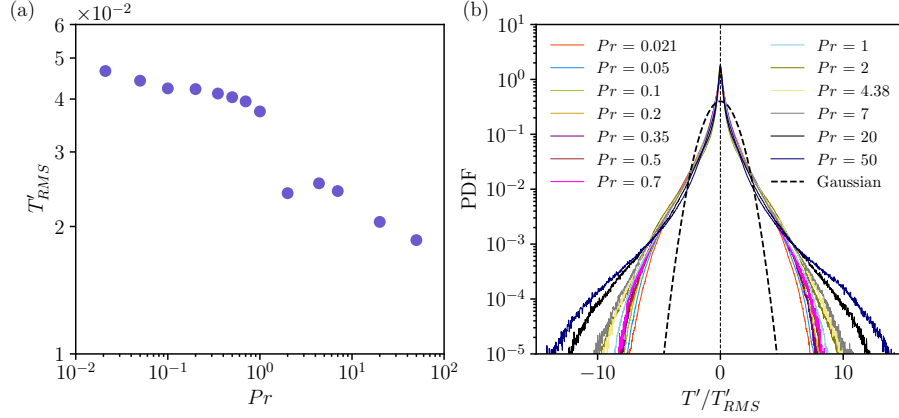


Figure 6: (a) Root mean square temperature fluctuation T'_{RMS} in the entire domain decreases with increasing Pr . An abrupt drop beyond $Pr = 1$ corresponds to a change in the flow morphology. (b) Probability density function of the normalized temperature fluctuation T'/T'_{RMS} is non-Gaussian for all the Prandtl numbers, and the departure from Gaussianity increases with increasing Pr .

4.1 Characteristics of temperature fluctuations

To characterize the flow fluctuations, we separate them from the mean fields using the Reynolds decomposition as [19]:

$$\mathbf{u}(\mathbf{x}, t) = \mathbf{U}(\mathbf{x}) + \mathbf{u}'(\mathbf{x}, t), \quad (6)$$

$$T(\mathbf{x}, t) = \Theta(\mathbf{x}) + T'(\mathbf{x}, t), \quad (7)$$

where $\mathbf{U}(\mathbf{x})$ and $\Theta(\mathbf{x})$ denote the velocity and temperature fields averaged over the entire simulation time, and primed quantities represent fluctuations. We define the RMS temperature fluctuation over the entire flow as

$$T'_{RMS} = \sqrt{\langle T'^2 \rangle_{V,t} - \langle T' \rangle_{V,t}^2} \quad (8)$$

and plot it as a function of Pr in figure 6(a). Note that $\langle T' \rangle_{V,t} = 0$ by definition (7). Figure 6(a) shows that the overall thermal fluctuation weakens as Pr increases; this weakening results from the decreasing TBL thickness with increasing Pr [20, 16], as shown in figure 5(a). Note that the highest contrasts in temperature are observed in regions that are occupied by thermal plumes, therefore the dominant contribution to T'_{RMS} comes from these regions. As the characteristic width of the thermal plumes scales as δ_T , the fraction of the flow occupied by the plumes decreases — and so is T'_{RMS} — as δ_T decreases, see also Pandey and Sreenivasan [16].

The probability density function (PDF) of T' in the entire flow domain is shown in figure 6(b). The distribution is non-Gaussian for all the Prandtl

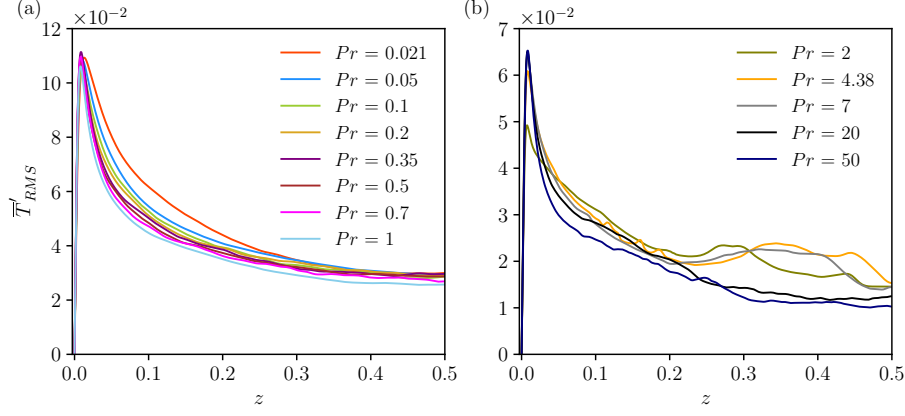


Figure 7: Vertical profiles of the RMS temperature fluctuation for (a) $Pr \leq 1$, and (b) $Pr > 1$. The profiles are additionally averaged over the lower and upper halves of the flow. The fluctuation increases rapidly in the vicinity of the plate and the peak fluctuation is observed inside the TBL region.

numbers and becomes wider as Pr increases. This suggests that the probability of finding the extreme relative fluctuations in the flow increases with increasing Pr , consistent with the observations in 3D convection [29, 20]. This is plausible because the thermal diffusion time becomes longer with increasing Pr that helps thermal structures with higher temperature anomalies retain their identity for a longer duration. To quantify the departure from a Gaussian distribution, we compute the kurtosis of temperature fluctuations [20] and find that it is significantly larger than three for all the Prandtl numbers. The kurtosis of T' increases from approximately 8 for $Pr = 0.021$ to 26 when Pr is increased to 50, which indicates that the temperature fluctuations become highly non-Gaussian when the Prandtl number is significantly larger than one.

To further explore the spatial distribution of temperature fluctuations, we compute the RMS fluctuation in a horizontal plane as

$$\bar{T}'_{RMS} = \sqrt{\langle T'^2 \rangle_{L,t} - \langle T' \rangle_{L,t}^2} \quad (9)$$

and show the profiles averaged over the upper and lower halves of the flow in figure 7. The fluctuation vanishes at the horizontal walls and increases steeply when moving away from them. Figure 7 exhibits that the peak fluctuation is observed in the vicinity of the wall and its location remains inside the TBL region. As discussed in Pandey [14], the increasing strength of fluctuation results from an increasing contrast between the temperatures of the plumes and the ambient fluid as distance from the plate increases. Beyond the TBL region, however, the mixing becomes efficient and the effective contrast decreases, causing a subsequent decline of $\bar{T}'_{RMS}$ beyond the peak. However, for $Pr = 2, 4.38$, and 7, the $\bar{T}'_{RMS}$ exhibits a broad low-amplitude peak in the bulk region. This secondary

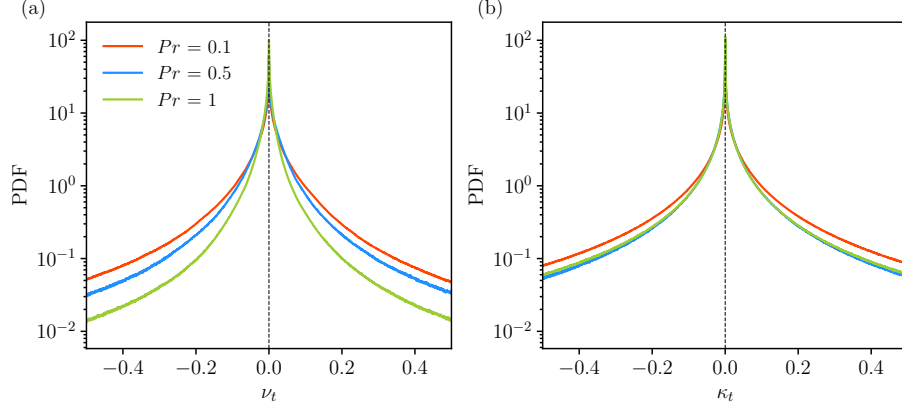


Figure 8: Probability density function of (a) the turbulent viscosity and (b) the turbulent thermal diffusivity in the entire flow domain. The PDFs are nearly symmetric, indicating that the positive and negative events are equally likely in convective flows. The tails of the PDF follow a stretched-exponential behaviour.

peak in the fluctuation profile arises due to the presence of strong corner flow structures for these parameters, see Sec. 3.1.

4.2 Turbulent Prandtl number

In the flux-gradient framework, turbulent viscosity and thermal diffusivity are estimated as

$$\nu_t = -\frac{\langle u'_x u'_z \rangle_t}{\partial U_x / \partial z} \quad \text{and} \quad \kappa_t = -\frac{\langle u'_z T' \rangle_t}{\partial \Theta / \partial z}. \quad (10)$$

Therefore, the presence of mean velocity and temperature gradients is crucial for applying the FG framework. As discussed in Sec. 1, this requirement, however, is generally not met in RBC due to a lack of the large-scale mean gradients in the flow [20]. We compute ν_t and κ_t at each point of the flow and look at their distributions by plotting their PDFs in figure 8. The PDFs, shown for three Prandtl numbers, are nearly symmetric and exhibit stretched-exponential tails. The symmetric nature of the PDFs suggest that it is difficult to estimate a mean turbulent Prandtl number over the entire flow [20, 1].

In the boundary layer region, however, mean gradients generally exist. We therefore show the vertical variation of the Reynolds stress and the turbulent heat flux in the TBL region in figure 9(a,b). The corresponding profiles of the mean velocity and temperature gradients are shown in figure 9(c,d). We observe that the turbulent fluxes and the corresponding gradients are of opposite signs in the region $0 \leq z \leq 2\delta_T$, thus yielding positive turbulent diffusivities ν_t and κ_t . To avoid clutter, we show profiles only for three Prandtl numbers in figure 9 but flows with $Pr \leq 1$ in this work exhibit qualitatively similar variations. As

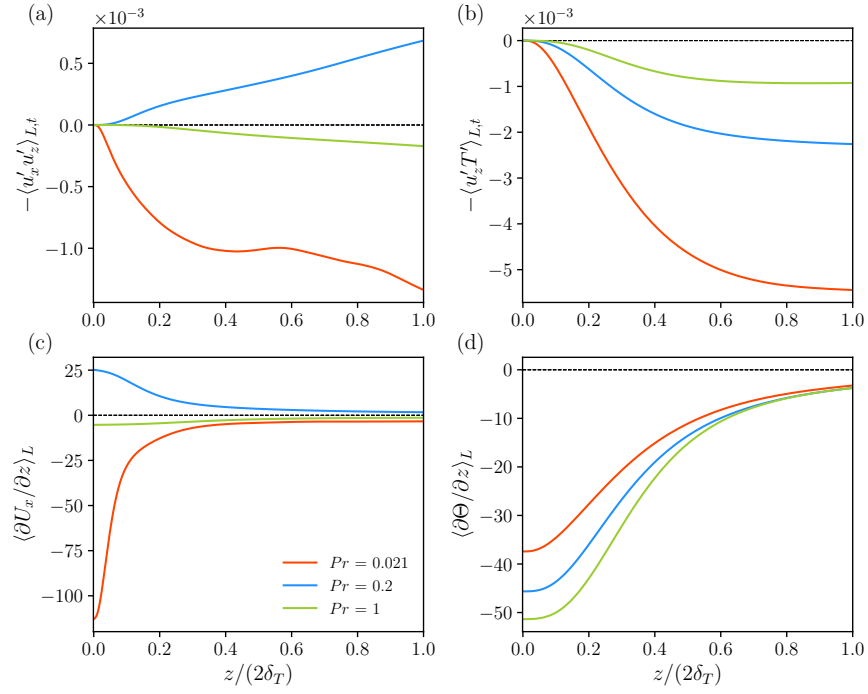


Figure 9: Horizontally-averaged turbulent fluxes (top) and the corresponding mean gradients (bottom) are of opposite signs in the TBL region near the bottom plate. These profiles yield positive turbulent diffusivities in the near-wall region, as shown in figure 10.

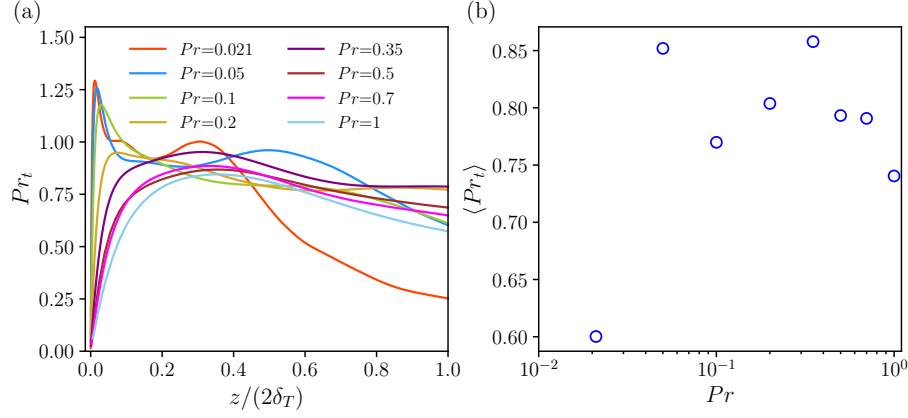


Figure 10: (a) Vertical profiles of Pr_t in the TBL region vary rapidly in the vicinity of the plate and gradually after exhibiting a maximum. The profiles are averaged over the top and bottom regions. (b) Pr_t averaged in the region $0.2 \leq z/(2\delta_T) \leq 1$ as a function of Pr varies from 0.60 to 0.85 without any systematic trend.

discussed in Sec. 3, the presence of strong corner flow structures for $1 < Pr \lesssim 10$ hinders the development of a mean velocity gradient, and a change in the sign of the mean gradient causes ν_t to diverge locally. Therefore, we compute the turbulent Prandtl number for flows only with $Pr \leq 1$.

Using the vertical profiles shown in figure 9, we compute the local turbulent Prandtl number in each horizontal plane

$$Pr_t(z) = \frac{\nu_t(z)}{\kappa_t(z)} = \frac{\langle u'_x u'_z \rangle_{L,t}}{\langle u'_z T' \rangle_{L,t}} \frac{\langle \partial \Theta / \partial z \rangle_L}{\langle \partial U_x / \partial z \rangle_L}, \quad (11)$$

and plot it in the TBL region in figure 10(a). As one moves away from the plate, $Pr_t(z)$ rises rapidly in its vicinity, achieves a maximum, and starts to decline gradually thereafter. This rapid growth however differs from a nearly constant $Pr_t(z)$ observed in the near-wall region of 3D RBC [34].

To investigate this differing trend, we look at the variations of $\nu_t(z)$ and $\kappa_t(z)$ in figure 11. We plot the normalized turbulent viscosity $(\nu_t/\nu)\xi^{-3}$ and thermal diffusivity $(\kappa_t/\kappa)\xi^{-3}$ as a function of ξ for $Pr = 0.05$ (figure 11(a)) and $Pr = 0.7$ (figure 11(b)). The normalized thermal diffusivity profile (red) is nearly flat up to $\xi \approx 0.1$, suggesting that κ_t scales as z^3 in this region. This is consistent with the proposed variation of κ_t in the near-wall region [30], and observations in 3D [34] as well as 2D flows [14]. In contrast, the normalized viscosity profiles (green) exhibit an increasing trend in the near-wall region. Thus, ν_t does not follow a z^3 scaling in the present 2D configuration and, instead, increases more rapidly in the vicinity of the horizontal plate. However, as ν_t has been observed to scale as z^3 in a cylindrical cell for various Ra and Pr [34], a differing behaviour

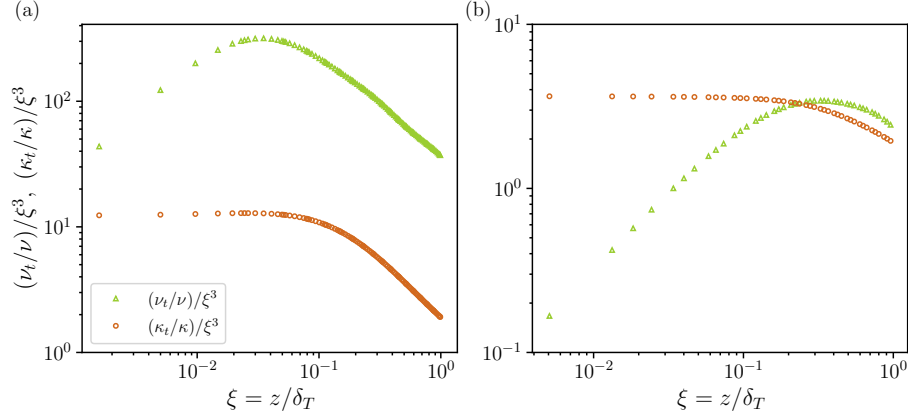


Figure 11: Vertical profiles of ν_t/ν and κ_t/κ normalized by ξ^3 in the bottom thermal boundary layer region for (a) $Pr = 0.05$ and (b) $Pr = 0.7$. Turbulent thermal diffusivity varies as ξ^3 , whereas turbulent viscosity exhibits a faster increase than ξ^3 in the vicinity of the plate.

in 2D indicates that the characteristics of the near-wall velocity field is probably different in 2D turbulent convection. This is in line with the findings from our recent study on the properties of viscous boundary layer, where we observed that the velocity distribution in 2D at higher Rayleigh numbers differs significantly from that in 3D [31].

Beyond the peak in the profile of $Pr_t(z)$ in figure 10(a), the variation is relatively weaker. Therefore, we calculate a mean turbulent Prandtl number $\langle Pr_t \rangle$ by averaging $Pr_t(z)$ in the region $0.2 \leq z/(2\delta_T) \leq 1$. The resulting $\langle Pr_t \rangle$ is plotted as a function of Pr in figure 10(b), which shows that the mean turbulent Prandtl number varies between 0.60 and 0.85 when Pr is varied by nearly two orders of magnitude. The figure further exhibits that it is difficult to infer any particular trend in the $Pr_t(Pr)$ curve. This is in line with the findings of [10] who observed nearly unchanging Pr_t in isotropically forced turbulence with imposed mean gradients. However, the trend detected in the present configuration differs from an increasing trend of Pr_t with decreasing Pr observed in the TBL region of 3D RBC [34], as well as with Pr_t estimated in the bulk region using other models [20, 19, 1].

5 CONCLUSIONS

Variation of fluid properties affects the flow behaviour. In RBC, this variation is achieved through the variation of the molecular Prandtl number. Here, we studied the influence of Prandtl number on the convective properties, such as the global flow organization and transports of heat and momentum. We considered a 2D layer of $\Gamma = 1$ imposed to a strong thermal driving ($Ra = 10^9$), and

varied Pr by more than three orders of magnitude. We found that the mean flow organization evolves as Pr is changed from low to high values. Further, the global transports scale differently in the low- and high- Pr regimes: $Nu \sim Pr^{0.08}$ and $Re \sim Pr^{-0.95}$ for low Prandtl numbers, whereas Nu varies weakly around a mean and $Re \sim Pr^{-0.87}$ for higher Prandtl numbers. To understand the trend of Nu with respect to Ra and Pr , we utilized the machine learning methods and trained a random forest regressor. We found that the regressor captures the variation of Nu in 2D RBC quite well and predicts Nusselt numbers that agree closely with those obtained from DNS.

We characterized the fluctuations of velocity and temperature fields by computing Pr_t using the FG framework, and found that the averaged turbulent Prandtl number in a horizontal plane varies rapidly in the vicinity of the top and bottom walls. The mean Pr_t within the TBL region exhibits no specific trend with respect to Pr and fluctuates between 0.6 and 0.85 for $0.021 \leq Pr \leq 1$. This differs from a decreasing trend with increasing Pr observed in 3D RBC [34]. To dig deeper, we looked at the variations of turbulent diffusivities and found that, similar to that in 3D, κ_t varies as z^3 near the horizontal walls. However, ν_t increases faster than z^3 , which differs from the corresponding 3D behaviour. Further exploration is therefore needed to understand these differing trends in 2D and 3D convective flows, which is currently underway and will be reported elsewhere.

Our findings suggest that the flow properties are altered in intricate ways when the flow is constrained to a vertical plane and, therefore, we need to be cautious when drawing inferences about the turbulent flows using 2D models [31].

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